19A 1 AMINES AND THEIR PREPARATIONS

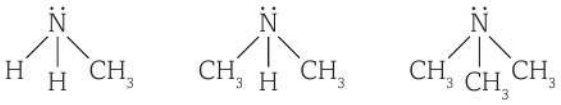
SPECIFICATION REFERENCE
19.1 PART | 19.4(i) | 19.4(ii) | 19.5

LEARNING OBJECTIVES

- Understand the nomenclature of amines and be able to draw their structural, displayed and skeletal formulae.
- Understand, in terms of reagents and general reaction conditions, the preparation of primary aliphatic amines:
  (i) from halogenoalkanes
  (ii) by the reduction of nitriles.
- Know the preparation of aromatic amines by the reduction of aromatic nitro-compounds using tin and concentrated hydrochloric acid.

INTRODUCTION TO AMINES

These nitrogen-containing compounds have some similarities with ammonia. The three bonding pairs of electrons around nitrogen are distributed in a trigonal pyramidal shape. The nitrogen atom has a lone pair of electrons and three bonds to one or more alkyl groups. If there is one alkyl group, the amine is classed as primary. If there are two, then it is secondary. With three it is tertiary. **Fig A** shows the structures of the simplest amine in each of these classes.



▲ **fig A** These are the simplest examples of primary, secondary and tertiary amines.

In this book we will focus on primary amines. Methylamine is a gas at room temperature, ethylamine has a boiling temperature of 17 °C, and propylamine and butylamine are volatile liquids. This list of names shows the naming system used – the suffix is *-amine* and the usual codes methyl, ethyl, propyl, butyl etc. are used.

We will also meet the simplest aromatic amine, $C_6H_5NH_2$, which is phenylamine.

Amines occur widely in nature, and many drugs (both legal and illegal) are amines, including the group of compounds called amphetamines.



▲ **fig B** Some cold and flu remedies contain amines.

NOMENCLATURE

We learned about amines in **Topic 10 (Book 1: IAS)**, but only as the product of reactions between halogenoalkanes and ammonia.

Table A shows the structural formulae and names of some examples of amines.

STRUCTURAL FORMULA	NAME
CH_3NH_2	methylamine
$CH_3CH_2NH_2$	ethylamine
$CH_3CH_2CH_2NH_2$	propylamine
$CH_3CH_2CH_2CH_2NH_2$	butylamine
$C_6H_5NH_2$	phenylamine

table A

DISPLAYED AND SKELETAL FORMULAE OF AMINES

Table B shows the displayed and skeletal formulae of the amines in **table A**.

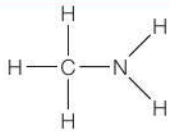
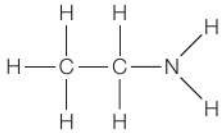
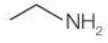
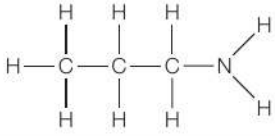
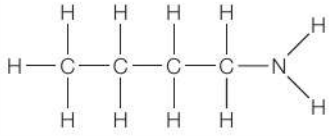
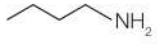
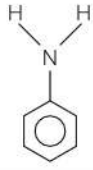
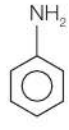
NAME	DISPLAYED FORMULA	SKELETAL FORMULA
methylamine		
ethylamine		
propylamine		
butylamine		
phenylamine		

table B

PREPARATION OF ALIPHATIC AMINES

There are two main ways of making primary aliphatic amines, starting from halogenoalkanes or from nitriles.

PREPARATION FROM HALOGENOALKANES

The method involves heating a halogenoalkane with ammonia. Because ammonia is a gas, this must be done under pressure and in a sealed container. Alternatively, the halogenoalkane can be mixed with concentrated aqueous ammonia. The equation for the preparation of methylamine is:

$$CH_3Cl + NH_3 \rightarrow CH_3NH_2 + HCl$$

The reaction involves nucleophilic attack by the lone pair of electrons of ammonia on the electron-deficient carbon atom in the halogenoalkane. Notice that the amine formed also has a nitrogen atom containing a lone pair of electrons. This means that it could also act as a nucleophile, competing with ammonia in the attack on the halogenoalkane. This would result in the reaction:

$$CH_3Cl + CH_3NH_2 \rightarrow (CH_3)_2NH + HCl$$

The product of this reaction is a secondary amine, in this case dimethylamine.

To prevent such unwanted side-reactions occurring, or at least reduce the chances of them happening, the ammonia is used in excess, so that it outnumbers the molecules of primary amine formed. Some of the excess ammonia reacts with the acidic hydrogen chloride formed, so a better equation to represent the preparation is:

$$CH_3Cl + 2NH_3 \rightarrow CH_3NH_2 + NH_4Cl$$

PREPARATION FROM NITRILES

Nitriles can be reduced to primary amines by reduction, using the reducing agent lithium tetrahydridoaluminate, which you may remember from **Topic 15** can also be used to reduce carbonyl compounds. As before, the reactants are mixed in dry ether, to ensure that there is no water that could affect the reaction. The equation for the reduction of ethanenitrile is:

$$CH_3CN + 4[H] \rightarrow CH_3CH_2NH_2$$

[H] represents hydrogen atoms produced by the reagent in a way that you do not need to know.

EXAM HINT

Although you can use [H] for the reducing agent, you must ensure the equation is balanced by **writing** the appropriate number. In this case it is 4[H].

PREPARATION OF AROMATIC AMINES

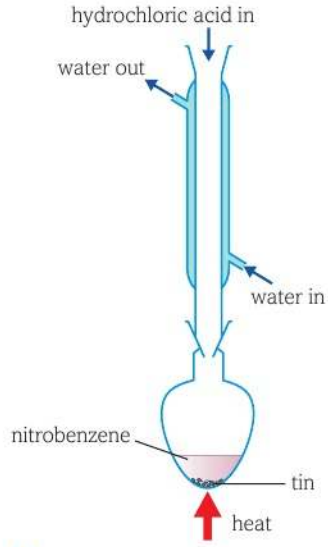
A specific method is used to prepare aromatic amines, especially phenylamine, which is made by the reduction of nitrobenzene. In this method, the reducing agent is tin mixed with concentrated hydrochloric acid, and the reaction mixture is heated under reflux (**fig C**). The reduction is achieved partly through the oxidation of tin to tin(II) ions and tin(IV) ions, and partly through the hydrogen produced in the reaction between tin and the acid. To make things simple we will represent the reducing agent by [H]. The equation for the reaction is:

$$C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$$

As with other amines, phenylamine is basic and will react with the acid present to form the phenylammonium ion, but this can easily be converted into phenylamine by adding an alkali such as sodium hydroxide solution:

$$C_6H_5NH_3^+ + OH^- \rightarrow C_6H_5NH_2 + H_2O$$

In the 19th century, phenylamine was called aniline, and was the source of large numbers of dyes used mainly for clothing. It is still used in the manufacture of synthetic indigo, most familiar as the colour of blue denim jeans. Nowadays, much phenylamine is used in the manufacture of polymers and pharmaceuticals.



▲ **fig C** Apparatus suitable for converting nitrobenzene into phenylamine.

LEARNING TIP

Note that in the name used for a nitrile, the code indicating the number of carbon atoms includes the C in the CN group, so $CH_3CH_2CH_2CN$ is butanenitrile, not propanenitrile.

CHECKPOINT

1. Write an equation for the preparation of propylamine starting from:
   (a) a halogenoalkane
   (b) a nitrile.
2. Explain why another organic product would be formed in 1(a) if the two reactants were used in a 1:1 ratio.

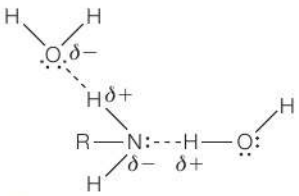
SKILLS DECISION-MAKING, PROBLEM-SOLVING

19A 2 ACID–BASE REACTIONS OF AMINES

SPECIFICATION REFERENCE 19.2(i) 19.2(ii) 19.3

LEARNING OBJECTIVES

- Understand the reactions of primary aliphatic amines (using butylamine as an example) and aromatic amines (using phenylamine as an example) with:
  (i) water to form an alkaline solution
  (ii) acids to form salts.
- Understand that amines are miscible with water as a result of hydrogen bonding, and the reasons for the difference in basicity between ammonia, primary aliphatic amines and primary aromatic amines.



▲ **fig A** The diagram shows the formation of hydrogen bonds between a primary amine and water.

REACTIONS WITH WATER

The first few members of the homologous series of primary aliphatic amines are completely miscible (capable of mixing) with water. However, as the hydrocarbon part of the molecule becomes proportionately larger, the solubility decreases.

They dissolve in water because they can form hydrogen bonds with water molecules (see **fig A**).

Phenylamine is only slightly soluble in water.

As well as dissolving in water, amines also react slightly with water to form alkaline solutions. Compare the equations using methylamine and ammonia:

$$CH_3NH_2 + H_2O \rightleftharpoons CH_3NH_3^+ + OH^-$$

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

You can see the similarity – this is because the nitrogen atom in both molecules can use its lone pair of electrons to form a dative bond with the hydrogen of a water molecule. The cation in the first equation is the methylammonium ion – think of it as an ammonium ion in which one hydrogen has been replaced by a methyl group. So, how do amines and ammonia compare as bases? What is the position of equilibrium in these reactions?

COMPARING BASICITIES

The **basicity** (basic strength) of a base can be quantified using the constant K_b or the constant pK_b. The pK_b of water at 298 K is 7.00, and any value lower than 7.00 indicates a basicity greater than that of water. Water is equally good as a base and as an acid. **Table A** shows the pK_b values at 298 K for ammonia and some amines.

LEARNING TIP

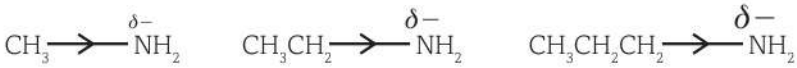
Note the big difference in basicity between aliphatic amines and aromatic amines and the small differences in basicity between different aliphatic amines.

NAME	FORMULA	pK_b
ammonia	NH_3	4.75
methylamine	CH_3NH_2	3.36
ethylamine	$CH_3CH_2NH_2$	3.27
propylamine	$CH_3CH_2CH_2NH_2$	3.16
phenylamine	$C_6H_5NH_2$	9.38

table A

What these values show is that methylamine is a stronger base than ammonia, and that extending the hydrocarbon chain causes further, but smaller, increases in basicity. Phenylamine is a much weaker base than any of the aliphatic amines, and is a weaker base than even water.

Methylamine is a stronger base than ammonia because the methyl group is electron-releasing, and so has an increased electron density on nitrogen compared with ammonia. The ethyl and propyl groups are only slightly more electron-releasing than methyl, so their effects are very similar. **Fig B** shows the situation.

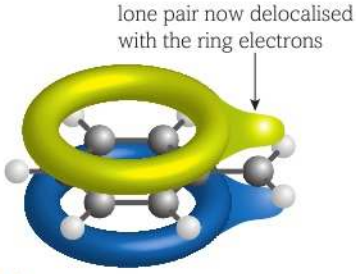


▲ **fig B** The diagram represents the slight increase in the electron-releasing effect of alkyl groups, causing a slight increase in electron density on the nitrogen atoms.

The equation for the reaction of butylamine with water is

$$CH_3CH_2CH_2CH_2NH_2 + H_2O \rightleftharpoons CH_3CH_2CH_2CH_2NH_3^+ + OH^-$$

The situation with phenylamine is very different. Think back to the diagram in **Section 18A.4** showing that in phenol one of the lone pairs of electrons on oxygen is incorporated into the delocalised electrons in the benzene ring. A similar thing happens with phenylamine, and this makes the nitrogen less electron-rich and the lone pair of electrons less available for donating to the hydrogen of a water molecule. **Fig C** shows how the delocalised electrons now extend above and below the nitrogen. So, as the original lone pair of electrons on nitrogen is now part of the delocalised system, it is less available for bonding to the hydrogen of a water molecule.



▲ **fig C** The electrons from the lone pair on the nitrogen have joined with the delocalised electrons on the benzene ring.

REACTIONS WITH ACIDS

Even though the amines vary in their basicity, they all react with strong acids to form ionic salts. Sample equations are:

$CH_3NH_2 + HNO_3 \rightarrow CH_3NH_3^+NO_3^-$
methylamine — methylammonium nitrate

$CH_3CH_2CH_2CH_2NH_2 + HCl \rightarrow CH_3CH_2CH_2CH_2NH_3^+Cl^-$
butylamine — butylammonium chloride

$C_6H_5NH_2 + HCl \rightarrow C_6H_5NH_3^+Cl^-$
phenylamine — phenylammonium chloride

CHECKPOINT

1. Write an equation for the reaction of ethylamine with:
   (a) water
   (b) sulfuric acid.
2. Predict, with a reason, whether dimethylamine has a lower or higher basicity than methylamine.

SKILLS CREATIVITY

SUBJECT VOCABULARY

basicity the extent to which a base can donate a lone pair of electrons to the hydrogen atom of a water molecule

19A 3 OTHER REACTIONS OF AMINES

SPECIFICATION REFERENCE
19.2(iii) 19.2(iv) 19.2(v) 19.6

LEARNING OBJECTIVES

- Understand the reactions of primary aliphatic amines (using butylamine as an example) and aromatic amines (using phenylamine as an example) with:
  (i) halogenoalkanes
  (ii) ethanoyl chloride
  (iii) copper(II) ions to form a complex ion.
- Be able to describe the reaction of aromatic amines with nitrous acid to form benzenediazonium ions, followed by a coupling reaction with phenol to form a dye.

LEARNING TIP

Addition–elimination reactions are also called condensation reactions.

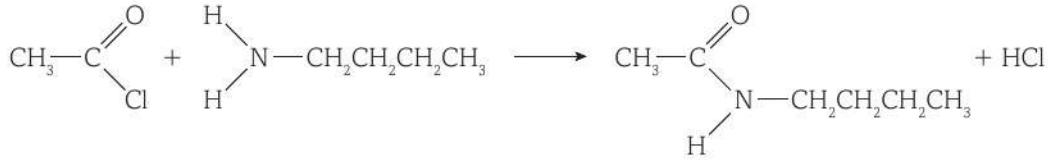
REACTIONS WITH ETHANOYL CHLORIDE

You do not need to know the mechanisms of these reactions. However, the name of the mechanism will help you understand how the reactions occur. The reaction type is **addition–elimination**, which means that the two molecules join together, and then a small molecule is eliminated – in these examples, hydrogen chloride. The organic product contains an new functional group – amide – in which a carbonyl group is next to an NH group.

The equation for the reaction of phenylamine with ethanoyl chloride is:

$$CH_3COCl + C_6H_5NH_2 \rightarrow CH_3CONHC_6H_5 + HCl$$

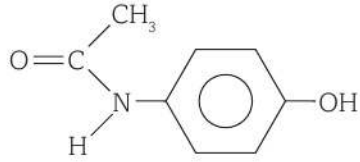
The equation for the reaction between ethanoyl chloride and butylamine is:



We will meet amides in a later section, but we can introduce the naming system here. The most common amide is ethanamide, and the product of the above reaction can be thought of as the result of replacing one of the hydrogens in the NH_2 group by the butyl group. Its name contains two words – the first is the alkyl group from the amine and the second indicates the number of carbon atoms in the original acyl chloride. As the butyl group is attached via a nitrogen atom, *N* is used as the locant. The name is *N*-butylethanamide.

PARACETAMOL

This is one of the most common pharmaceuticals used to relieve the symptoms of fever and pain. It is manufactured in a sequence of reactions, one of which is an addition–elimination reaction. Its structure is:



You can see how paracetamol got its name. The 'para' part indicates that the two groups attached to the benzene ring are at opposite ends of the ring. The 'acetam' part comes from the old name for ethanamide, which used to be called acetamide. Finally, the 'ol' part indicates the presence of a hydroxyl group.

REACTIONS WITH HALOGENOALKANES

You do not need to know the mechanisms of these reactions. However, you can see that the two would react together because a halogenoalkane contains an electron-deficient carbon atom and

an amine contains an electron-rich nitrogen atom. Using general formulae, the reaction can be represented as:

$$R'NH_2 + R''X \rightarrow R'NHR'' + HX$$

where R′ is the alkyl group in the amine and R″ is the alkyl group in the halogenoalkane. The reaction is another example of substitution. The organic product is a secondary amine and the inorganic product is a hydrogen halide, often hydrogen chloride.

The equation for the reaction between phenylamine and chloroethane is:

$$C_6H_5NH_2 + CH_3CH_2Cl \rightarrow C_6H_5NHCH_2CH_3 + HCl$$

An example equation using butylamine and chloroethane is:

$$CH_3CH_2CH_2CH_2NH_2 + CH_3CH_2Cl \rightarrow CH_3CH_2CH_2CH_2NHCH_2CH_3 + HCl$$

Notice that the organic product also contains an electron-rich nitrogen atom, so it can also react with chloroethane. The equation for this further reaction is:

$$CH_3CH_2CH_2CH_2NHCH_2CH_3 + CH_3CH_2Cl \rightarrow CH_3CH_2CH_2CH_2N(CH_2CH_3)_2 + HCl$$

The organic product of this reaction is a tertiary amine.

Once again, the tertiary amine contains a nitrogen atom with a lone pair of electrons, which can also react with the halogenoalkanes. The equation for this further reaction is:

$$CH_3CH_2CH_2CH_2N(CH_2CH_3)_2 + CH_3CH_2Cl \rightarrow CH_3CH_2CH_2CH_2N^+(CH_2CH_3)_3Cl^-$$

Note that the equation is different this time – HCl is not formed because this would require the loss of H from the nitrogen of the organic reactant, which a tertiary amine does not have. The product is an ionic compound related to ammonium chloride ($NH_4^+Cl^-$), except that all the hydrogens in the ammonium ion have been replaced by alkyl groups. It is known as a quaternary ammonium salt.

Reaction with halogenoalkanes does not seem to be a good method of preparation, as it is likely that a mixture of similar products will be formed. If there is an excess of the halogenoalkane, then the sequence of all four reactions is more likely to occur. Producing a quaternary ammonium salt in this sequence is what happens in industry. These compounds have many uses, and are usually found in fabric softeners.

REACTIONS WITH COPPER(II) IONS

You will remember from **Topic 17** that ammonia can act as a lone pair donor in its reactions with transition metal ions. This is the overall equation for its reaction with hexaaquacopper(II) ions:

$$[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$$

Amines also have a lone pair of electrons on nitrogen, so can take part in similar reactions. The observations are the same as with ammonia – first of all a pale blue precipitate forms, then with excess butylamine the precipitate dissolves to give a deep blue solution. The equations for these reactions look complicated, but the only difference is the use of $CH_3CH_2CH_2CH_2NH_2$ in place of NH_3.

LEARNING TIP

Be careful not to confuse amines, ammines and amides, all of which appear in this section.

Formation of the pale blue precipitate:

$$[Cu(H_2O)_6]^{2+} + 2CH_3CH_2CH_2CH_2NH_2 \rightarrow [Cu(H_2O)_4(OH)_2] + 2CH_3CH_2CH_2CH_2NH_3^+$$

Formation of the deep blue solution:

$$[Cu(H_2O)_4(OH)_2] + 4CH_3CH_2CH_2CH_2NH_2 \rightarrow [Cu(CH_3CH_2CH_2CH_2NH_2)_4(H_2O)_2]^{2+} + 2H_2O + 2OH^-$$

With phenylamine, the overall equation to form the complex is:

$$[Cu(H_2O)_6]^{2+} + 4C_6H_5NH_2 \rightarrow [Cu(C_6H_5NH_2)_4(H_2O)_2]^{2+} + 4H_2O$$

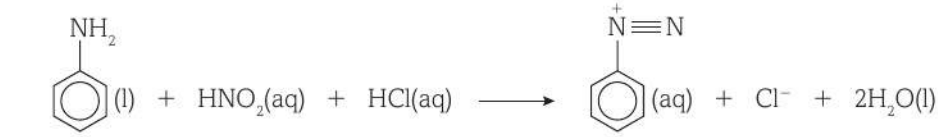
FORMATION OF AZO DYES

REACTION OF PHENYLAMINE WITH NITROUS ACID

Nitrous acid is a very unstable compound. It exists only in aqueous solution and decomposes at room temperature. This means that any reactions involving nitrous acid have to be carried out under very carefully controlled conditions. The nitrous acid must be prepared in situ by mixing ice-cold solutions of sodium nitrite and dilute hydrochloric acid. The equation for the reaction is:

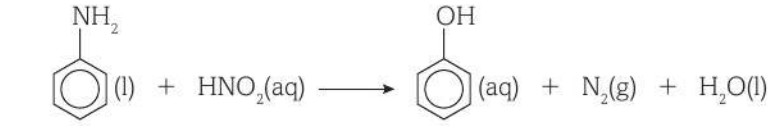
$$NaNO_2(aq) + HCl(aq) \rightarrow NaCl(aq) + HNO_2(aq)$$

If an aromatic amine, such as phenylamine, is then added to this reaction mixture, a diazonium ion is formed:



phenylamine benzendiazonium ion

The reaction vessel must be kept in an ice-water mixture because the reaction must be carried out at 5 °C or lower. If the temperature rises above 5 °C, phenol is formed:

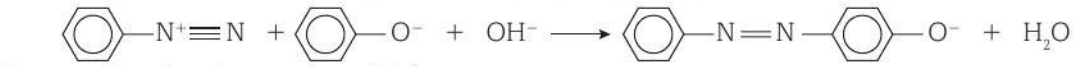


phenylamine phenol

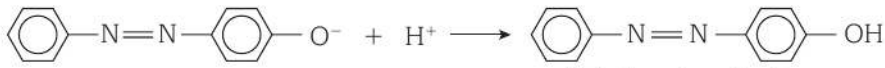
REACTION OF DIAZONIUM IONS WITH PHENOLS

Diazonium ions carry a positive charge and therefore can act as strong electrophiles. For example, the benzenediazonium ion reacts readily in cold, alkaline solution with both aromatic phenols and amines.

The benzenediazonium ion reacts with an alkaline solution of phenol to produce, after acidification with dilute hydrochloric acid, a compound called (4-hydroxyphenyl)azobenzene:



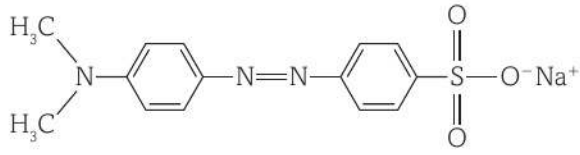
benzenediazonium ion phenoxide ion



(4-hydroxyphenyl)azobenzene

This reaction is called a **coupling reaction**. The compound formed is an energetically stable, yellow azo dye. Its stability comes from extensive delocalisation of electrons via the nitrogen-to-nitrogen double bond. If you examine the structure you can see how the double and single bonds alternate. Systems of alternating double and single bonds are called **conjugated systems**. Where extensive conjugation occurs, absorption of light takes place, so the compound is coloured.

The dye (4-hydroxyphenyl)azobenzene is just one of the wide range of dyes that can be made from aromatic amines and other arenes. These are known as azo dyes. They are very stable, so they do not fade. Another example is the acid–base indicator methyl orange, which has the structure:



CHECKPOINT

SKILLS DECISION-MAKING

1. Write an equation for the reaction between propanoyl chloride and propylamine.
2. Draw the structure of the quaternary ammonium salt formed when ethylamine reacts with 1-bromopropane.

SUBJECT VOCABULARY

addition–elimination reaction when two molecules join together, followed by the loss of a small molecule
coupling reaction a reaction in which two organic molecules or ions join together to form one new molecule
conjugated system where single and double bonds alternate, allowing the electrons in the p-orbitals of the atoms to overlap and form a delocalised electron cloud

19A 4 AMIDES AND POLYAMIDES

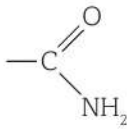
SPECIFICATION REFERENCE
19.1 19.7 19.8(i) PART 19.8(ii) 19.9 PART 19.10

LEARNING OBJECTIVES

- Understand the nomenclature of amides and be able to draw their structural, displayed and skeletal formulae.
- Understand that amides can be prepared from acyl chlorides.
- Be able to describe:
  (i) condensation polymerisation for the formation of polyamides such as nylon
  (ii) addition polymerisation, including poly(propenamide) and poly(ethenol).
- Be able to draw the structural formulae of the repeat units of polyamides, poly(propenamide) and poly(ethenol).
- Be able to comment on the physical properties of polyamides and the solubility in water of the addition polymer poly(ethenol) in terms of hydrogen bonding, including soluble laundry bags or liquid-detergent capsules (liquitabs).

AMIDES

We learned about amides in previous sections as the products of reactions. Amides have a functional group consisting of a carbonyl group joined to an amino group as shown.



EXAM HINT

You might expect the amide functional group to show basic character, but it does not. The carbonyl group alters the chemical character of the NH_2 group, just like the carbonyl group alters the chemical character of the OH group in carboxylic acids.

As with other compounds in which two functional groups share the same carbon, amides have properties that are different from both carbonyl compounds and amines. They are solids (except for methanamide, which is a liquid). Also, the lower aliphatic amides are soluble in water because they contain two electronegative atoms and polar bonds. This means that they can form hydrogen bonds with water. The carbon atom is very electron-deficient because it is joined to both nitrogen and oxygen.

NOMENCLATURE

Table A shows the structural formulae and names of some amides.

STRUCTURAL FORMULA	NAME
$HCONH_2$	methanamide
CH_3CONH_2	ethanamide
$CH_3CH_2CONH_2$	propanamide
$CH_3CH_2CH_2CONH_2$	butanamide

table A

DISPLAYED AND SKELETAL FORMULAE OF AMIDES

Table B shows the displayed and skeletal formulae of the amides in **table A**.

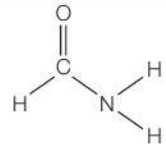
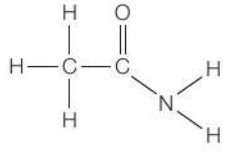
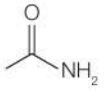
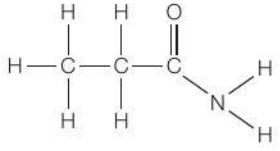
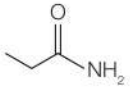
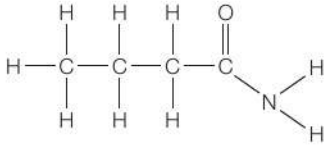
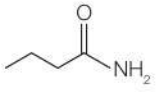
NAME	DISPLAYED FORMULA	SKELETAL FORMULA
methanamide		
ethanamide		
propanamide		
butanamide		

table B

PREPARATION OF AMIDES

A convenient way to prepare amides in the laboratory is by mixing an acyl chloride with concentrated aqueous ammonia. Although you do not need to know the mechanisms of these reactions, it helps to understand the reaction if we refer to the mechanism. The lone pair of electrons on the nitrogen of the ammonia molecule is strongly attracted to the electron-deficient carbon atom of the acyl chloride. The chlorine of acyl chloride combines with one of the hydrogen atoms of ammonia to form hydrogen chloride, which appears as misty fumes. The equation for the reaction between propanoyl chloride and ammonia is:

$$CH_3CH_2COCl + NH_3 \rightarrow CH_3CH_2CONH_2 + HCl$$

Note that ammonia is basic and that the inorganic product is acidic, so there will be a reaction between the two molecules:

$NH_3 + HCl \rightarrow NH_4Cl$

So, a better overall equation for the reaction, combining these equations, is:

$CH_3CH_2COCl + 2NH_3 \rightarrow CH_3CH_2CONH_2 + NH_4Cl$

POLYAMIDES



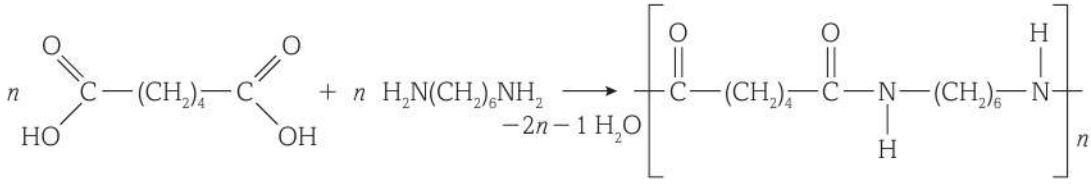
▲ **fig A** Balloon fabric is made of nylon.

In **Topic 15** we learned about polyesters as examples of polymers formed by condensation polymerisation reactions. The formation of these polymers needs two monomers – a dicarboxylic acid and a diol. The formation of polyamides also needs two monomers – a dicarboxylic acid and a diamine.

The simplest examples of a dicarboxylic acid and a diamine are HOOCCOOH and $H_2NCH_2NH_2$. You can see how these molecules can react together – one has acidic groups and the other has basic groups, so they can react with each other to form a CONH group with the elimination of H_2O.

NYLON

Nylon is almost certainly the most familiar example of a polyamide. Both of the monomers used in the production of its most common form contain six carbon atoms. Common examples of these monomers are hexanedioic acid (known as adipic acid in the polymer industry) and hexane-1,6-diamine (also known as 1,6-diaminohexane). Both of these monomers have the same number of CH_2 groups with reactive groups at each end of the chain. The formation of this polymer is shown below:



hexanedioic acid 1,6-diaminohexane nylon 6,6

There are different types of nylon, and this example is described as nylon 6,6 because both of the monomers contain six carbon atoms.

LEARNING TIP

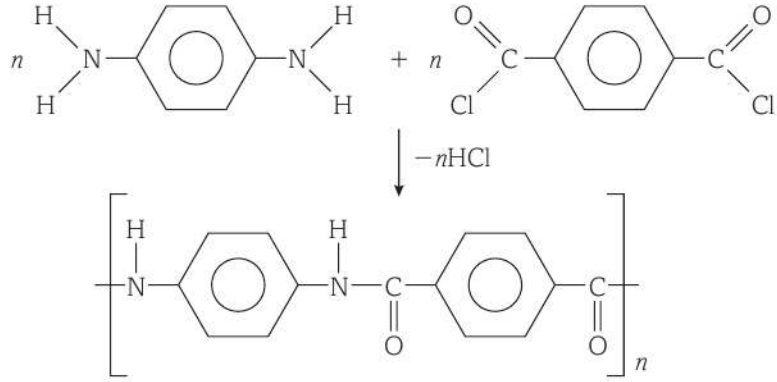
In the formation of a polyamide, the small molecule formed is water (if a dicarboxylic acid is used) or hydrogen chloride (if a dioyl chloride is used).



▲ **fig B** Kevlar® is used to make bulletproof vests.

KEVLAR®

If benzene rings take the place of the CH_2 groups in the monomers used to make nylon 6,6, and a dioyl chloride is used in place of a dicarboxylic acid, then the monomers and the polymer structure can be shown like this:



This polymer produced is known as Kevlar® and has very many uses, the most familiar of which is as body armour (bulletproof and stabproof vests). However, the list of applications continues to grow, as a web search will reveal.

PROPERTIES OF POLYAMIDES

The majority of polyamides (nylons) tend to be semi-crystalline. They are generally very tough materials with good thermal and chemical resistance. Polyamides tend to absorb moisture from

their surroundings. This absorption continues until equilibrium is reached. In general, the impact resistance and flexibility of polyamides tends to increase with moisture content, while the strength and stiffness decrease.

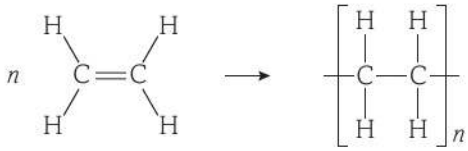
The strong bonds in polyamides make the polyamide chains themselves strong. Polyamides such as nylon can be made into strong fibres for making clothing, fishing lines and carpets. Kevlar® is also a polyamide.

Polyamide film (cling film) is used for food packaging because of its toughness and because gas molecules do not pass through it easily. Polyamide film is also used for 'boil-in-the-bag' food packaging because of its high temperature resistance.

Polyamides tend to provide good resistance to most chemicals. However, they can be attacked by strong acids and alkalis. This is because they contain the amide link (peptide link), which is hydrolysed by acids and alkalis (see **Section 19A.6**).

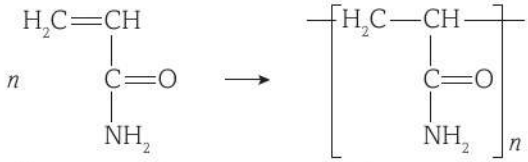
POLY(PROPENAMIDE)

In **Topic 5 (Book 1: IAS)** you learned about addition polymerisation. This is when many molecules (monomers) with double bonds join together to form a long-chain molecule (polymer). In this reaction, one of the bonds in the double bond is broken and the monomers are linked together by single bonds. The most common example of this is the formation of poly(ethene) from ethene:



ethene poly(ethene)

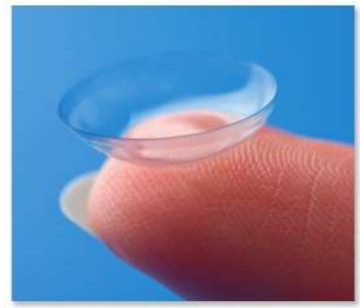
Poly(propenamide) is an addition polymer. It is formed by polymerising 2-propenamide (sometimes called acrylamide):



2-propenamide poly(propenamide)

The polymer is used as a thickener and as a filler in facial surgery. It is also used in water treatment and papermaking.

The polymer chains in poly(propenamide) can be easily cross-linked with other chains. The resulting polymer with cross-chains produces a thick gel that has a larger capacity to absorb water. It can be used for making soft contact lenses (**fig C**).

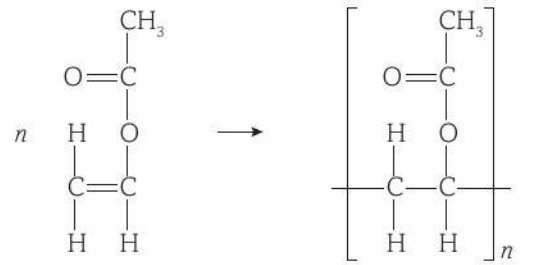


▲ **fig C** Soft contact lenses are made of poly(propenamide).

POLY(ETHENOL)

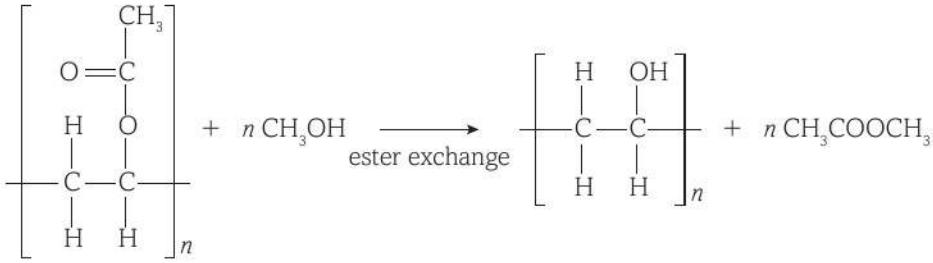
Poly(ethenol) is another addition polymer. It is sometimes called poly(vinyl alcohol). It is not manufactured in the usual way by directly polymerising a monomer. Instead it is made in two stages.

The first stage is the polymerisation of ethenyl ethanoate:



ethenyl ethanoate poly(ethenyl ethanoate)

In the second stage, poly(ethenyl ethanoate) is reacted with methanol to form poly(ethenol) and methyl ethanoate, in a process called ester exchange:



poly(ethenyl ethanoate) poly(ethenol)

The amount of ester exchange can be controlled by altering the temperature.

Poly(ethenol) molecules contain many OH groups. These groups can form hydrogen bonds with water, and so the polymer can be made to be soluble in water. The exact solubility of the final polymer depends on the percentage of ester groups that have been replaced by OH groups. The solubility can range from insoluble, to soluble in hot/warm water, to soluble in cold water.

Poly(ethenol) is used to make disposable laundry bags for use in hospitals. The laundry bags are soluble in water. Dirty bedding and clothes that may be covered in microorganisms from patients are put into the laundry bag. The bag is then placed into a washing machine without the hospital workers touching the dirty fabrics. During washing, the bag dissolves completely and the fabrics are washed clean.

Poly(ethenol) is also used to make liquid-detergent capsules (liquitabs) that contain measured quantities of detergent for use in washing machines and dishwashers. The bags dissolve in the water.

CHECKPOINT

SKILLS INTELLECTUAL INTEREST AND CURIOSITY

1. Write the overall equation for the formation of ethanamide from an acyl chloride.
2. Write an equation for the formation of Kevlar® starting from a diamine and a dicarboxylic acid.

19A 5 AMINO ACIDS

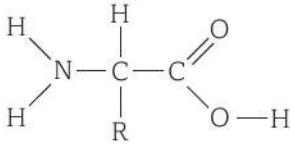
SPECIFICATION REFERENCE
19.1 PART 19.11(i) 19.11(ii)

LEARNING OBJECTIVES

- Understand the nomenclature of amino acids and be able to draw their structural, displayed and skeletal formulae.
- Be able to describe experiments to investigate the characteristic behaviour of amino acids limited to:
  (i) acidity and basicity and the formation of zwitterions
  (ii) effect of aqueous solutions on plane-polarised monochromatic light.

WHAT ARE AMINO ACIDS?

The name suggests the presence of an amino group and a carboxylic acid group. Unlike amides, these two groups are separated by a carbon atom and so retain most of their typical properties. Below is the general displayed formula of an amino acid.



When R (the symbol for an alkyl group) is only a hydrogen, the amino acid is the simplest one, glycine. Over 20 amino acids are found in humans. Some of these are synthesised in the body, and others must be included in the diet. Proteins are made from amino acids, and these will be considered in the next section. Although R is normally used to represent an alkyl group, in the case of amino acids it can represent a more complex structure. **Table A** shows information about some important amino acids.

NAME	ABBREVIATION	STRUCTURE	ISOELECTRIC POINT
alanine	ala	$H_2N-CH(CH_3)-COOH$	6.0
cysteine	cys	$H_2N-CH(CH_2-SH)-COOH$	5.1
glutamic acid	glu	$H_2N-CH(CH_2-CH_2-COOH)-COOH$	3.2
glycine	gly	H_2N-CH_2-COOH	6.0
lysine	lys	$H_2N-CH(CH_2-CH_2-CH_2-CH_2-NH_2)-COOH$	9.7

table A

There are 20 naturally occurring amino acids. They are often referred to as 2-amino acids because the NH_2 is attached to the second carbon atom in the chain, counting the C in COOH as carbon 1. Note that most amino acids have one NH_2 group and one COOH group, although some have two of one and one of the other.

NOMENCLATURE

Although amino acids are usually referred to by their common names, such as glycine, H_2NCH_2COOH, and alanine, $H_2NCH(CH_3)COOH$, they also have systematic names using the rules defined by IUPAC. They are named as amine derivatives of carboxylic acids, using the prefix *amino* to indicate the presence of the amine group, NH_2. A locant is used to indicate the position of the amine group in the carbon chain of the carboxylic acid.

Table B shows the structural formulae and names of some amino acids. The 3-amino acids are not naturally occurring.

STRUCTURAL FORMULA	NAME
H_2NCH_2COOH	2-aminoethanoic acid
$CH_3CH(NH_2)COOH$	2-aminopropanoic acid
$H_2NCH_2CH_2COOH$	3-aminopropanoic acid
$CH_3CH(NH_2)CH_2COOH$	3-aminobutanoic acid
$C_6H_5CH(NH_2)COOH$	2-aminophenylethanoic acid

table B

DISPLAYED AND SKELETAL FORMULAE OF AMINES

Table C shows the displayed and skeletal formulae of the amines in **table B**.

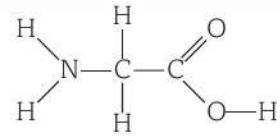
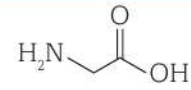
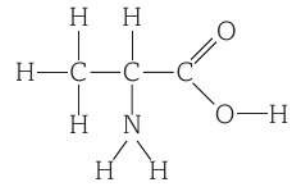
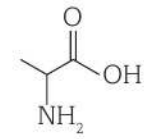
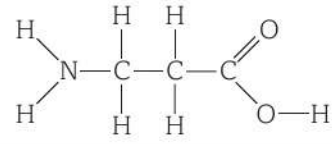
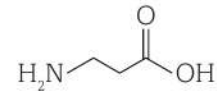
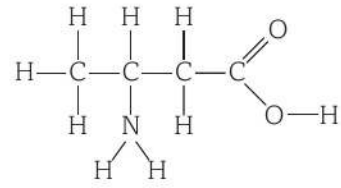
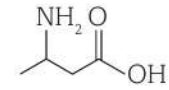
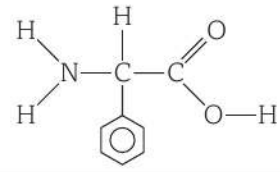
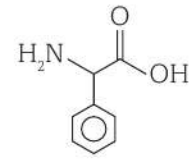
NAME	DISPLAYED FORMULA	SKELETAL FORMULA
2-aminoethanoic acid		
2-aminopropanoic acid		
3-aminopropanoic acid		
3-aminobutanoic acid		
2-aminophenylethanoic acid		

table C

ACIDIC AND BASIC PROPERTIES

The values of the **isoelectric points** of the amino acids have been included to help you understand their acid–base character. They are all soluble in water, and you can imagine that one of two reactions might occur. The molecule could act as a base and form an alkaline solution in a reaction like this:

$$H_2N{-}CHR{-}COOH + H_2O \rightleftharpoons H_3N^+{-}CHR{-}COOH + OH^-$$

Alternatively, it could act as an acid and form an acidic solution in a reaction like this:

$$H_2N—CHR—COOH + H_2O \rightleftharpoons H_2N—CHR—COO^- + H_3O^+$$

There is a third alternative – an H^+ ion could transfer from the COOH group to the NH_2 group in a reaction like this:

$$H_2N—CHR—COOH \rightleftharpoons H_3N^+—CHR—COO^-$$

The product of this reaction is electrically neutral because it has a positive charge and a negative charge that balance each other. Such species are called **zwitterions**. *Zwitter* is the German word for hybrid, meaning a cross between two things and having the characteristics of both.

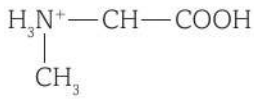
Now to the isoelectric point. This is the pH at which the zwitterion exists in aqueous solution. A low isoelectric point indicates that the molecule is predominantly acidic (glutamic acid has a value of 3.2), while a high value indicates that it is predominantly basic (lysine has a value of 9.7). You can see that these values are related to the numbers of NH_2 and COOH groups in the molecule (glutamic acid has an extra carboxylic acid group and lysine has an extra amino group).

EXAM HINT

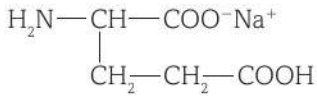
There is a link here to **Topic 14** on acid–base equilibria. You might like to think about how proteins could be involved in buffering pH.

SALT FORMATION

All amino acids can form salts with acids and bases. For example, alanine can react with acids to form this protonated structure:



Glutamic acid can react with sodium hydroxide to form three possible salts. This is because there are two COOH groups, so either of them can react, or both can react. One salt has the structure:



You may have heard of this salt – its name is monosodium glutamate and is often used as a flavour enhancer in food.



▲ **fig A** Monosodium glutamate can be used as a flavour enhancer in cooking.

OPTICAL ACTIVITY

Almost all 2-amino acids contain a chiral centre (the C of the CH group), and so are optically active. The exception is glycine, which has a CH_2 group instead. Aqueous solutions of the enantiomers rotate the plane of polarisation of plane-polarised light – some enantiomers are dextrorotatory (+) and others are laevorotatory (−). If an amino acid is synthesised in the laboratory, then a racemic mixture is formed.

LEARNING TIP

Practice writing equations for different amino acids to show how they can react as both acids and bases.

CHECKPOINT

1. Draw the structure of glycine in a solution with:
   (a) a pH of 6.0
   (b) a very low pH
   (c) a very high pH.
2. Draw structures to show how the two enantiomers of cysteine are related to each other.

SUBJECT VOCABULARY

isoelectric point (of an amino acid) the pH of an aqueous solution in which it is neutral

zwitterion a molecule containing positive and negative charges but which has no overall charge

19A 6 PEPTIDES AND PROTEINS

SPECIFICATION REFERENCE
19.9 PART 19.11(iii)

LEARNING OBJECTIVES

- Be able to draw the structural formulae of the repeat units of proteins.
- Be able to describe experiments to investigate the characteristic behaviour of amino acids in the formation of peptide bonds by condensation polymerisation.

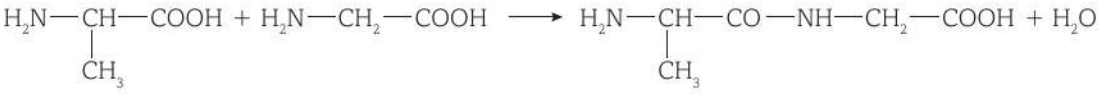
WHAT IS A PEPTIDE?

When two amino acid molecules react together, an acid–base reaction occurs. The OH of the COOH group combines with one of the H atoms of the NH_2 group to form water. This is a condensation reaction, in which the two amino acids are joined together by an amide group (CO—NH). The bond that forms is known as a **peptide bond**, and the organic product is a dipeptide.

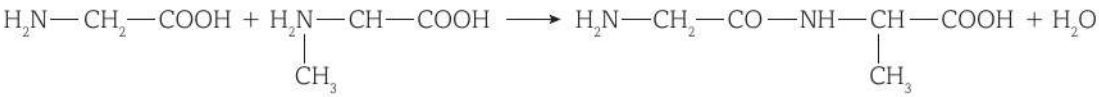
MORE THAN ONE DIPEPTIDE

When two amino acids combine together to form a dipeptide, there are always two possibilities – sometimes more. For example, when glycine reacts with alanine, the OH could be lost from either molecule, as shown below.

alanine + glycine:



glycine + alanine:



TRIPEPTIDES

When one molecule of each of three different amino acids reacts together to form a tripeptide, the six possibilities can be summarised using the three-letter abbreviations shown in the table in the previous section. They are:

ala–cys–glu ala–glu–cys cys–ala–glu cys–glu–ala glu–ala–cys glu–cys–ala

If you work out the possibilities in which two or more molecules of the same amino acid in these three react to form a tripeptide you will find that there are many more!

POLYPEPTIDES AND PROTEINS

Polypeptides and **proteins** are formed by condensation polymerisation of many amino acids. The main difference between a long-chain polypeptide and a protein is that proteins have further levels to their structures. These are to do with the way in which the polypeptide chains interact with each other in three dimensions, to give secondary, tertiary and quaternary structures, but these are beyond the aims of this book.

ANALYSING PROTEINS

Many proteins have very large molar masses and complex structures. In recent decades, more and more proteins have been analysed. As several million different proteins are thought to exist, this work is still continuing but far from complete. **Table A** shows three examples of proteins, the first two of which have vital roles in the human body.

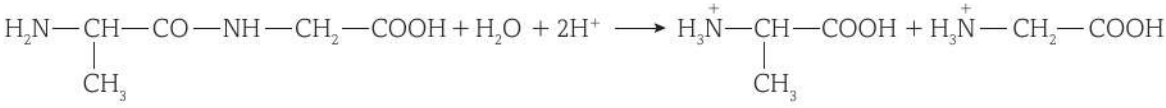
PROTEIN	WHERE FOUND	APPROXIMATE MOLAR MASS/g mol^{-1}	APPROXIMATE NUMBER OF AMINO ACIDS
insulin	pancreas	5700	51
haemoglobin	blood	66 000	574
urease	soya beans	480 000	4500

table A

The first step in discovering the structure of a protein is to find which amino acids are present. The second is to find the order in which they occur in the polypeptide chains.

HYDROLYSING PROTEINS

The polypeptide chains in a protein can be broken down into their individual amino acids by prolonged heating with concentrated hydrochloric acid. This breaks the peptide bonds between the amino acids, although, because of the strongly acidic conditions, all the amino acids formed will have their NH_2 groups protonated as $^+NH_3$ groups. You can see this process more easily in the hydrolysis of a dipeptide. As an example we will use the dipeptide formed between alanine and glycine, referred to earlier in this section. The equation for this hydrolysis is:



You can see that this hydrolysis reaction is the reverse of the original dipeptide formation, except that the two NH_2 groups are protonated, which explains the inclusion of the two H^+ ions in the equation.

USING CHROMATOGRAPHY

The different types of chromatography are covered in **Topic 15**, and you already know about simple chromatography from your previous study of chemistry. A mixture of amino acids produced by hydrolysis of a protein can be spotted onto chromatography paper. Using a suitable solvent, the individual amino acids will rise to different heights during the experiment. As amino acids are colourless, the chromatogram can be sprayed with a developing agent so that the positions of the amino acids can be seen. You may read about the use of ninhydrin as a developing agent. Although this works well, ninhydrin is not normally used now because of its toxic nature.

However, once the positions of the amino acids have been established, their R_f values can be calculated, and so the individual amino acids can be identified. Working out how they are all joined together is rather more complicated and beyond the aims of this book.

Insulin is a protein with the molecular formula $C_{256}H_{381}N_{65}O_{79}S_6$. It is made up of a total of 51 amino acids, including 17 different ones. Its structure is shown in abbreviated form in **fig A**.



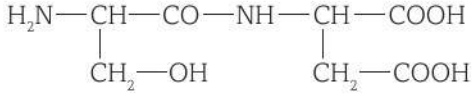
▲ **fig A** This diagram shows the sequence of amino acids in insulin from a sheep. You can see that there are actually two separate sequences of amino acids in which cysteine molecules are joined by disulfide bonds.

EXAM HINT

Remember that separating amino acids using chromatography depends on the relative solubilities of the amino acids in the mobile and stationary phases. It *does not* depend on the size of the amino acids (see **Topic 15**).

CHECKPOINT

1. Write the structures of the two dipeptides formed when alanine and cysteine react together.
2. Write the structures of the two amino acids formed when this dipeptide is hydrolysed by concentrated hydrochloric acid:



SKILLS PROBLEM-SOLVING

LEARNING TIP

Two possible dipeptides can form between two different amino acids. If one or both of the amino acids has more than one NH_2 or COOH group, then more possible dipeptides can be formed.

SUBJECT VOCABULARY

peptide bond the bond formed by a condensation reaction between the carbonyl group of one amino acid and the amino group of another amino acid

polypeptide a condensation polymer formed from many amino acids

protein a polypeptide that has folded into a specific shape in order to have a specific function